

Alexander Marshall | Physicist

1900 Glenacre Drive – Kernersville, North Carolina 27284

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🌐 philosophiaephysica.org • Royster Doctoral Fellow at UNC Chapel Hill

HR Department

Corporation

123 Pleasant Lane

12345 City, State

November 21, 2025

To Whom it May Concern,

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Sincerely,

Alexander Marshall

Attached: Curriculum Vitae

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Education

The University of North Carolina at Chapel Hill <i>Doctorate of Philosophy in Physics</i> Royster Doctoral Fellow	Chapel Hill, NC <i>Ongoing</i>
The University of North Carolina at Chapel Hill <i>Master's Degree in Physics</i>	Chapel Hill, NC <i>2025</i>
Wake Forest University <i>B.S. with Honors in Physics, B.A. in Philosophy, Minor in Math.</i> GPA 3.91 <i>Summa Cum Laude</i> Stamps Scholar	Winston-Salem, NC <i>2018-2022</i>
West Forsyth High School <i>High School Diploma</i> Junior Marshall Top 1% of graduating class	Clemmons, NC <i>2014-2018</i>

Academic Projects

Honors Thesis.....

Generating a 3-D Lattice of Photonic Spheres to Track Fluorescently Labelled Chromatin

Advisor: Dr. Keith Bonin

Committee: Dr. Keith Bonin, Dr. George Holzwarth, Dr. Stephen Baker

Description: Wrote a custom iterative routine to generate, test, and analyze hundreds of thousands of phase settings on a spatial light modulator, implementing the correction method of Dholakia et al. (2017) Used this routine to create 3D lattice of photonic spheres to be used for tracking labelled chromatin in irradiated cells.

Masters Thesis.....

Physiological Strains in Poroelastic Media Induce Asymmetric Transport.

Advisor: Richard Superfine

Thesis Committee: Dr. Richard Superfine, Dr. Greg M. Forrest, Dr. David Hill, Dr. Amy Oldenburg, Dr. Kerry Bloom

Description: Tracked quantum dots undergoing dynamic diffusion in poroelastic media using LSM with unprecedented temporal resolution. Live imaging during physiological strain reveals that large solute transport is asymmetric during dynamic strains, with advection yielding significantly more forward angular anisotropy during the relaxation stroke than the compression stroke. These results suggest single-particle effects could explain the macroscopic solute concentration enhancements reported by studies of poroelastic media under dynamic strain.

Doctoral Thesis.....

Anomalous Diffusion in Poroelastic Media

Advisor: Richard Superfine

Thesis Comittee: Dr. Richard Superfine, Dr. C.A. Forrest, Dr. David Hill, Dr. Amy Oldenburg, Dr. Ehssan Nazockdast

Description: Explore the role of static and dynamic strains in the transport of large solutes in various poroelastic media, including transcription factor search patterns in cellular nuclei exposed to dynamic strains.

Additional Research.....

Wake Forest Nanotech Center <i>Investigating Chalcogenide Crystals for Quantum Computing, David Carroll</i>	Winston-Salem, NC 2021
Wake Forest Medical School <i>Examining Neurochemical Addiction Mechanisms, Evgeny Budygin</i>	Winston-Salem, NC 2019-2021
Wake Forest Department of Anthropology <i>Cultural Impacts of Oenology in the Peleponnese, Karen Friederic</i>	Winston-Salem, NC 2019
Wake Forest Baptist Medical Center <i>Cancer and Postoperative Pain, Chris Peters</i>	Winston-Salem, NC 2018

Scholarships and Grants

2022-2027: Royster Doctoral Fellowship. Five-year fellowship awarded for academic merit.

2018-2022: Stamps Scholarship. Four-year full scholarship awarded for academic merit.

2019: Wake Forest Scholars Travel Grant. Awarded to fund anthropology project in Greece.

Publications

Technical skills

Advanced Coding: MATLAB, LabVIEW, Python, $\LaTeX$

Scripting: μ Manager, ImageJ

Data Analysis: Particle Tracking, Image Analysis

Laboratory Skills: Transmission Microscopy, Fluorescent Microscopy, Light Sheet Fluorescent Microscopy, Total Internal Reflection Microscopy, Optics, 4F Laser Systems, Spatial Light Modulator, Remote Focusing, Particle Tracking, Machine Vision, Atomic Force Microscopy

Talks

Generating a 3-D Lattice of Photonic Spheres <i>Symposium, Wake Forest University Physics Department</i>	Winston-Salem, NC May 2022
Diffusion in Strained Poroelastic Media <i>Thesis Proposal, UNC Chapel Hill Physics Department</i>	Chapel Hill, NC September 16, 2025
Strain Alters Anomalous Diffusion in Poroelastic Media. <i>Symposium, Elon University Physics Department</i>	Elon, NC October 1, 2025
Physiological Strains in Poroelastic Media Induce Asymmetric Transport. <i>Master's Defense, UNC Chapel Hill Physics Department</i>	Chapel Hill, NC November 17, 2025

Conferences

Undergraduate Research Day <i>Symposium, Wake Forest University URECA Center</i>	Winston-Salem, NC <i>September 2019</i>
Strain Driven Transport In Poroelastic Media <i>Triangle Cytoskeleton Meeting, Duke University</i>	Durham, NC <i>September 15, 2025</i>
Strain Driven Transport In Poroelastic Media <i>Triangle Cytoskeleton Meeting, NC Museum of Natural Sciences</i>	Raleigh, NC <i>September 23, 2024</i>
Probing Structural and Force Dynamics during Phagocytosis <i>Triangle Cytoskeleton Meeting, NC Museum of Natural Sciences</i>	Raleigh, NC <i>September 25, 2023</i>
Various topics <i>Annual Colloquium, UNC Applied Physical Sciences Department</i>	Chapel Hill, NC <i>2022-2025</i>
Strain Driven Transport In Poroelastic Media <i>8th International Soft Matter Conference, Soft Matter Association of the Americas</i>	Raleigh, NC <i>August 2024</i>
Strain Driven Transport In Poroelastic Media <i>Triangle Soft Matter Workshop, Duke University Physics Department</i>	Durham, NC <i>May 12, 2025</i>
Strain Driven Transport In Poroelastic Media <i>13th Carolina Biophysics Symposium, UNC Chapel Hill</i>	Chapel Hill, NC <i>November 6, 2025</i>
Strain Driven Transport In Poroelastic Media <i>Royster Global Conference, Royster Society of Fellows</i>	Chapel Hill, NC <i>October 21, 2025</i>
Strain Driven Transport In Poroelastic Media <i>Biophysical Society Meeting, Biophysical Society</i>	San Francisco, CA <i>February 21, 2026</i>

Teaching and Mentorship

2021-Present: Many semesters as a Teaching Assistant.

2018-Present: Private tutoring in Physics, Mathematics, Chemistry.

Coordinated a team of student researchers: 2023-Present

Outreach

Wake Forest Baptist Medical Center <i>Volunteer, over 400 accumulated hours</i>	Winston-Salem <i>2015–2017</i>
UNC APS Department <i>Open Flexure Microscope and Molecular Stamper</i> Constructing low cost microscopes for use in low-resource environments. Project has recieved multiple awards, including the UNC MakerFest Entrepreneurship Award.	Chapel Hill <i>2023-Present</i>
UNC Chapel Hill <i>UNC ScienceFest</i>	Chapel Hill <i>2023-Present</i>

Select Coursework

University of North Carolina at Chapel Hill	Chapel Hill, NC
Graduate Mathematical Methods	2022
Graduate Quantum Mechanics	2022

Graduate Classical Dynamics	2022
Graduate Electromagnetism	2022
Graduate Statistical Mechanics	2023
Optics	2024
Fluid Dynamics	2025
Biophysics	2025
Wake Forest Univeristy	Winston-Salem, NC

2022	<i>Partial Differential Equations</i>
2022	<i>Quantum Computing</i>

References

Richard Superfine, PhD: superfine@unc.edu	<i>Doctoral Advisor</i>
Keith Bonin, PhD: bonin@wfu.edu	<i>Undergraduate Research Advisor</i>
Evgeny Budygin, PhD, PharmD: budygin@gmail.com	<i>Undergraduate Research Mentor</i>

Languages

English: Fluent	<i>Native Speaker</i>
Modern Greek: Intermediate	<i>Conversational</i>
Classical Latin: Intermediate	<i>Several years of formal instruction</i>

Interests

-Piano	-Chess
-Long-Distance Running	-Backpacking